Abdalla Abdelmagid

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EDUCATION

Towson University, Towson, MD — *B.S. Computer Science* — May 2025
GPA: 3.8/4.0

- **Key Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Mobile App Development, Software Quality Assurance & Testing, Object-Oriented Programming, Web-Based Programming
- **Honors:** Cum Laude, 8x Dean's List

TECHNICAL SKILLS

Languages: Java, Swift, Python, JavaScript/TypeScript, SQL, C++, HTML/CSS
Frameworks & Libraries: Spring Boot, React, Node.js, Express, SwiftUI, JUnit
Databases & Tools: MySQL, MongoDB, Git, AWS, Docker, Firebase

EXPERIENCE

Independent iOS Developer — Remote
Self-Employed — Jan 2025 – Present

- Shipped 14 native iOS apps to the App Store, reaching over 10,000 downloads and maintaining a 4.6 average rating.
- Took apps through the entire lifecycle, from concept and Figma design to development, testing, and successful App Store launches.
- Monetized apps through subscriptions and in-app purchases, generating over $1000 in monthly recurring revenue.

Data Science Research Fellow — Baltimore, MD
National Science Foundation (NSF) — Sep 2024 – Dec 2024

- Analyzed Baltimore City crime/resource datasets with Python (pandas) and SQL, applying regression and clustering to uncover correlations.
- Built interactive Tableau dashboards and Matplotlib visualizations to highlight key trends.
- Presented insights and policy recommendations to NSF and university stakeholders.

Software Engineer Intern — Remote
Revature — May 2024 – Aug 2024

- Engineered 6 RESTful API endpoints in Java/Spring Boot with JWT authentication, securing application data for 1,000+ users.
- Optimized database queries with Spring Data JPA, reducing latency by 30% and improving platform responsiveness.
- Authored 15+ unit/integration tests with JUnit 5 and Mockito, uncovering recurring bugs and preventing crashes.

Software Engineer Intern — Columbia, MD
HQLabs — May 2023 – Aug 2023

- Developed responsive front-end features using React and TypeScript for a real-time financial transaction monitoring dashboard.
- Built and integrated RESTful services to process financial transaction data, improving application reliability and performance.
- Collaborated in an Agile development environment using Git for version control and CI/CD pipelines to streamline software delivery.

PROJECTS

University Marketplace — Feb 2025

- Built secure campus marketplace (MERN, Firebase Auth) enabling verified .edu sign-ups, real-time chat, and Stripe-powered payments.
- Designed flexible MongoDB schemas and optimized queries to ensure smooth loading and filtering of marketplace items.

nVolve — Event Discovery App — Nov 2024

- Developed SwiftUI app displaying geolocated events; used Alamofire + Codable for REST API consumption and CoreLocation for GPS services.
- Implemented a MapKit-based interface with filtering and favorites features to help users find and bookmark nearby events.

Movie Recommender Web App — May 2024

- Collaborated in an Agile team of 4 to design a responsive UI (HTML/CSS/JS) and build a RESTful backend (Java, Spring) interfacing with MySQL and OpenAI GPT-3 to suggest films.
- Implemented JWT-based authentication and deployed the application in a Dockerized environment for public access.